

Maxwell Kwesi Mac-Ocloo

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EDUCATION

Michigan State University, MSU

Doctor of Philosophy in Biostatistics

East Lansing, Michigan

August 2023 – **Expected May 2028**

The University of Texas at El Paso, UTEP

Master of Science in Statistics and Data Science

El Paso, Texas

August 2021 – May 2023

University of Energy and Natural Resources, UENR

Bachelor of Science in Statistics

Sunyani, Ghana

August 2015 – July 2019

RELEVANT COURSEWORK

Statistical Programming, Statistical Data Mining, Statistical Consulting (analysis and design), Generative Models, Spline Regression, Linear Regression Model, Mixed Model, Survival Analysis, Causal Inference, Mediation Analysis, Semi-parametric and Non-parametric Models.

SKILLS AND COMPETENCIES

Software Development: SAS, R Studio, Python Programming, SPSS, STATA, Microsoft Office

Soft skills: Effective verbal, written communication and interpersonal skills, time management skills, fast learner, team player, strong analytical skill.

EXPERIENCE

Biostatistics Intern

Public Health Sciences, Henry Ford Health

May 2026 – August 2026

Detroit, Michigan

- Collaborated with investigators and interdisciplinary research teams to formulate research questions, select appropriate statistical methodologies, interpret findings, and translate statistical results into scientifically meaningful conclusions.
- Contributed to scientific manuscripts and research reports by developing statistical methods sections, interpreting analytical results, synthesizing relevant literature, and preparing results for peer-reviewed dissemination.
- Designed and implemented statistical analyses for observational clinical and public health studies addressing treatment adherence, behavioral health, social support, substance use, and survival outcomes.

Data Analyst

Department of Obstetrics, Gynecology and Reproductive Biology, MSU

June 2025 – April 2026

East Lansing, Michigan

- Leveraged R and SAS for advanced statistical modeling—causal inference, hazard modeling, trend analysis and survival analysis—to examine prognostic determinants of breast cancer mortality across racial groups.
- Conducted rigorous statistical analyses to evaluate racial disparities in breast cancer survival and assess changes in mortality patterns across diagnosis periods.
- Developed clear reports, publication-quality visualizations, and statistical summaries to communicate findings to faculty and research collaborators.

Graduate Research Assistant

Department of Epidemiology and Biostatistics, MSU

August 2023 – May 2025

East Lansing, Michigan

- Applied causal inference and missing data methodologies to clinical and epidemiological datasets using R and SAS.
- Developed and implemented statistical models (e.g., sequential mean models, regression, survival analysis) to evaluate treatment effects in observational studies using SAS.
- Collaborated with interdisciplinary teams to translate statistical findings into actionable insights for health research.

Data Science Intern

Lawrence Berkeley National Laboratory

June 2023 – August 2023

Berkeley, California

- Managed and analyzed large-scale high-performance computing datasets to optimize system efficiency.
- Derived and visualized performance metrics using Python, demonstrating scalability in big data environments.
- Explored and derived performance metrics from large datasets and visualized performance metrics using Python.

Graduate Teaching Assistant

August 2021 – May 2023

The University of Texas at El Paso

El Paso, Texas

- Developed and delivered lesson plans tailored to diverse learning needs, reinforcing core mathematical concepts and problem-solving skills.
- Administered, graded, and tracked weekly assignments and quizzes for 40 students to monitor academic progress and provide timely feedback.
- Tutored 55 students in Proportion and Algebra through structured and innovative instructional approaches.

PUBLICATIONS

- Das, J.; Beyaztas, B.H.; **Mac-Ocloo, M.K.**; Majumdar, A.; Mandal, A. Testing Equality of Multiple Population Means under Contaminated Normal Model Using the Density Power Divergence. *Entropy* 2022, 24, 1189.
- Asosega, K. A., Opoku-Ameyaw, K., Otoo, D., **Mac-Ocloo, M. K.**, & Ayinzoya, R. (2020). Comparative Sensitivity Performance of the Discriminant Function and Logistic Regression under Different Training and Test Samples for Predicting Birth Outcomes. *Asian Journal of Probability and Statistics*, 6(3), 47-60.

PRESENTATIONS

- Exploring Real-time Data Transfers from Stanford Linear Accelerator Center (SLAC) to National Energy Research Scientific Computing Center (NERSC).
- Variance component testing in generalized linear models with random effect by Xihong Lin.
- Association between Social Support and Treatment Adherence in Head and Neck Cancer Patients.

ACADEMIC HONORS

- Valedictorian, University of Energy and Natural Resources 2019 Congregation
- Best student award in the School of Sciences, University of Energy and Natural Resources 2019 Congregation
- Best student award in the Department of Mathematics and Statistics, University of Energy and Natural Resources 2019 Congregation.

LEADERSHIP AND SERVICE

Vice President

August 2025 – Present

Biostatisticians and Epidemiologists at Michigan State (BEAMS)

East Lansing, Michigan

- Support organizational leadership, strategic planning, and execution of initiatives focused on academic development, mentorship, professional growth, and community engagement.
- Collaborate with executive board members to coordinate programs, events, and activities that promote student success and professional development.

Vice President

April 2026 – Present

African Graduate Student Association (AGSA), MSU

East Lansing, Michigan

- Provide leadership and advocacy for African graduate students, supporting academic, professional, cultural, and social programming within the university community.
- Collaborate with student leaders and university stakeholders to organize events, strengthen community engagement, and promote an inclusive and supportive graduate student environment.